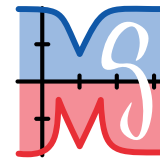


MoleGraph: The Open Gateway to Science

Empower your **STEM** education and Inquiry-Based Learning (IBL) with MoleGraph. This versatile **open-source DIY Probeware system**, powered by **Arduino NANO**, invites you to look inside the "black box" of scientific measurement. Build your **own instruments**, analyze real-time data via Bluetooth, and explore the principles of electronics. Designed for limitless creativity, the system is fully compatible with the **m-BITBEAM** construction system, allowing you to easily integrate sensors into custom experiments and robotic projects.



Name

List of sensors in this box (checkmarks)

ID	Sensor	Measured Quantities [Unit]	ID	Sensor	Measured Quantities [Unit]
1	Thermometer (DS18B20)	Temperature [°C]	19	Alcohol gas sensor (MQ-3)	Concentration [ppm], Concentration [%], Voltage [V]
2	Pressure + Thermometer + Hygrometer (BME280)	Pressure [Pa], Temperature [°C], Humidity [%], Altitude [m], Dew point [°C]	20	Particle/Dust sensor (DSM501)	Concentration 1 [ppm], Concentration 2 [ppm], Concentration 3 [ppm]
3	Accelerometer + Magnetometer (LSM303DLHC)	Acceleration [m/s ²], Magnetic Field Intensity [μT], Heading [°], Roll [°], Pitch [°]	21	Sound sensor (microphone KY-037)	RAW [], Voltage [V], Trigger []
4	IR Thermometer (MLX90614)	Temperature (Sensor) [°C], Temperature (Object) [°C]	22	DC Voltage sensor 0-25 V module	Voltage [V], RAW []
5	Pressure sensor (MPX5700DP)	Pressure [Pa]	23	AC/DC Current sensor 0-5 A (ACS712)	Current [A], RAW []
6	Force sensor (using HX711 module)	Force [N]	24	AC/DC Current sensor 0-30 A (ACS712)	Current [A], RAW []
7	ECG sensor (AD8232)	Voltage [V], Frequency [BPM], Trigger []	25	UV sensor (VEML6070)	UV Intensity []
8	Color sensor (TCS3200)	RAW_R [], RAW_G [], RAW_B []	26	Light sensor (BH1750 luxmeter)	Illuminance [lux]
9	UV sensor (GUVA-S12SD)	UV Index []	27	Thermometer (K-Type, MAX6675)	Temperature [°C]
10	pH sensor (E201 + pH-4502C)	pH [], Voltage [V], RAW [], Trigger []	28	Length sensor (Caliper)	Length [mm]
11	Conductivity sensor (TDS)	Conductivity [μS/cm], Salinity (ppm) [ppm], RAW [], Voltage [V]	29	Combustible Gas (MQ-2) LPG, CNG	Concentration [ppm], Concentration [%], Voltage [V]
12	Voltage + Current sensor (using HX711 module)	Voltage [V], Current [A]	30	ORP sensor (DFRobot SEN0165)	RAW [], Voltage [mV], Trigger []
13	Light sensor (photoresistor module)	Illuminance [RAW], Voltage [V], Trigger []	31	Turbidity sensor (TS-300B)	Turbidity [NTU], Voltage [V], Trigger []
14	CO ₂ gas sensor (MH-Z16)	Concentration [ppm], Concentration [%]	32	Photogate (810H)	Pulse duration [s], Period [s], Frequency [Hz], Pulse count []
15	O ₂ gas sensor (ME2-O2)	Concentration [ppm], Concentration [%]	33	Output/input prototype breadboard	
16	Motion/distance US sensor (SR-HC04)	Distance [m], Velocity [m/s], Acceleration [m/s ²]	34	Spirometer (MPXV7002DP)	Volume [l], RAW [], Voltage [V], Trigger []
17	Heart rate sensor (simple module)	RAW [], Frequency [BPM], Trigger []	35	Radioactivity (RadiationD v1.1)	Radioactivity [cpm]
18	Magnetic field sensor (Hall module KY-024)	RAW [], Voltage [V], Trigger []	36	Motion/distance laser sensor (VL53L0X)	Distance [m], Velocity [m/s], Acceleration [m/s ²]

<http://www.e-mole.cz/diy/molegraph>
<http://www.e-mole.cz/diy/molegraph-shield>

Applications for platforms: MS Windows, macOS, Linux, Android

Download on GitHub



Designed & Coded in Czech Republic

MoleGraph v4.3

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